

Import Libraries

```
import numpy as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving retail_sales_dataset (1) (1).csv to retail_sales_dataset (1) (1) (1).csv

```
import pandas as pd
df = pd.read_csv('retail_sales_dataset (1) (1).csv')
df.head()
```

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30

```
print(df.shape)
print(df.columns)
df.info()
df.describe()
```

```
(1000, 9)
Index(['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age',
      'Product Category', 'Quantity', 'Price per Unit', 'Total Amount'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Transaction ID         1000 non-null  int64
1   Date                   1000 non-null  object
2   Customer ID           1000 non-null  object
3   Gender                 1000 non-null  object
4   Age                    1000 non-null  int64
5   Product Category      1000 non-null  object
6   Quantity               1000 non-null  int64
7   Price per Unit         1000 non-null  int64
8   Total Amount           1000 non-null  int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
```

	Transaction ID	Age	Quantity	Price per Unit	Total Amount
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	500.500000	41.39200	2.514000	179.890000	456.000000
std	288.819436	13.68143	1.132734	189.681356	559.997632
min	1.000000	18.00000	1.000000	25.000000	25.000000
25%	250.750000	29.00000	1.000000	30.000000	60.000000
50%	500.500000	42.00000	3.000000	50.000000	135.000000
75%	750.250000	53.00000	4.000000	300.000000	900.000000
max	1000.000000	64.00000	4.000000	500.000000	2000.000000

```
df.isnull().sum()
df = df.dropna()
```

```
x = df[['Age', 'Quantity']]
y = df['Total Amount']
```

```
X_train,X_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

Train Model

```
model=LinearRegression()
model.fit(X_train,y_train)
```

▼ LinearRegression ⓘ ?

LinearRegression()

```
X = df[['Age', 'Quantity']]
y = df['Total Amount']
```

Train Test split

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LinearRegression
```

Prediction

```
Y_pred=model.predict(X_test)
```

Evaluation

```
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
import numpy as np
print(mean_absolute_error(y_test,Y_pred))
print(mean_squared_error(y_test,Y_pred))
print(np.sqrt(mean_squared_error(y_test,Y_pred)))
print(r2_score(y_test,Y_pred))
```

```
446.56304849838125
273012.21057137026
522.5057038649151
0.06735569178126377
```

Predict New File

```
new_location = pd.DataFrame([[4, 3]], columns=['Age', 'Quantity'])
pred = model.predict(new_location)
print(pred)
```

```
[656.55761697]
```

Actual Predicted

```
result = pd.DataFrame({'Actual Total Amount': y_test.values, 'Predicted Total Amount': Y_pred})
```

```
import matplotlib.pyplot as plt
plt.scatter(y_test, Y_pred)
plt.xlabel('Actual Total Amount')
plt.ylabel('Predicted Total Amount')
plt.grid(True)
plt.show()
```

